

Dexter Fernandes

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AI Software Engineer

AI Software Engineer with 3+ years of experience optimising and deploying deep learning systems on accelerated hardware under strict latency, reliability, and resource constraints. Strong C++17 and PyTorch background with hands-on experience translating model performance analysis into actionable optimisation guidance for cross-functional teams and customers. Trusted contributor in safety-critical environments, combining systems thinking, measurement-driven iteration, and clear technical communication to enable robust AI deployments.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Designed and deployed real-time CNN perception pipelines on edge accelerators using PyTorch and OpenVINO, achieving a 15% F1 improvement with deterministic 24/7 operation under harsh industrial conditions.
- Developed a repeatable model optimisation workflow (baseline → FP16 → validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.
- Applied FP16 optimisation on Intel VPU hardware, meeting strict latency and power budgets for safety-critical inference without compromising reliability.
- Built benchmarking and profiling harnesses to compare accuracy, latency, and stability pre- and post-optimisation, supporting measurement-driven iteration and regression detection.
- Diagnosed and resolved production failures through on-site analysis and structured validation checklists, reducing crash frequency by 67% in live video pipelines.
- Refactored legacy monolith into Kafka-driven, Dockerised microservices, improving fault isolation and increasing MTBF by 35% across edge deployments.
- Documented deployment guardrails, failure modes, and recovery procedures, reducing cognitive load for engineers operating and maintaining fielded systems.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy CNN-based models for real-time object detection, pose estimation, and tracking, reducing false positives by 33% and achieving sub-50 ms inference latency on edge hardware.
- Defined and standardised FP16/INT8 optimisation paths (including QAT, pruning, and layer fusion), reducing model size by 50% while preserving accuracy under tight memory and latency constraints.
- Re-architected the embedded inference pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Mentored engineers on optimisation trade-offs, deployment best practices, and performance debugging, improving delivery consistency across AI, software, and hardware teams.
- Built automated labelling, dataset versioning, and performance-tracking pipelines, repurposing detection models for annotation and cutting manual labelling effort by 80%, accelerating iteration cycles.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system integrating LiDAR and radiological data for the teleoperation of the Clearpath Husky UGV development platform.
- Implemented ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes and interfaces integrating perception outputs into a broader robotic system, managing sensor topics, tf2 frames, and message synchronisation for downstream localisation and navigation.
- Integrated perception pipelines into ROS-based robotic workflows, validating end-to-end data flow from sensors to SLAM and navigation components in Gazebo prior to deployment.
- Simulated and validated 2D/3D mapping and navigation workflows in Gazebo and ROS2 prior to real-world deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

PROJECTS

ROS2 SLAM & Nav2 Simulation

Oct 2025

- Built a simulated mobile robot integrating 3D LiDAR and RGB camera data in Gazebo Harmonic to perform real-time SLAM and autonomous navigation using ROS 2 Jazzy.
- Applied slam_toolbox and Nav2 for 2D/3D mapping, localisation, and waypoint-based navigation with optimised Ceres backends.
- Processed multi-sensor data streams for spatial alignment and tested mapping accuracy across diverse simulated environments.
- Developed modular ROS 2 launch files and parameterised configurations for flexible experimentation with perception and navigation pipelines.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

SKILLS

- **Programming & Development:** Bash, C++17, CMake, colcon, Git, Linux, Python, rclcpp, rclpy, ROS1 (Melodic/Noetic), ROS2 (Humble/Foxy/Jazzy)
- **Deep Learning and Optimisation:** Bayesian Optimisation, Layer fusion, ONNX, OpenVINO, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Structured pruning, TensorFlow, TensorRT
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Perception & SLAM:** Convolutional Neural Networks, Data Augmentation, Eigen, LiDAR SLAM, NumPy, Object Detection & Tracking, OpenCV, Point Cloud Library, ROS2 image_transport, Scikit-Image
- **Cloud, Edge & MLOps:** Apache Kafka, Docker, DVC (Data Version Control), MLFlow